

Evaluating the Value of Topology-Aware Features for Cyber-Physical Grid Monitoring: Detection, Localization, and Real-Time Feasibility

Yagiz Sarapli

Department of Electrical Engineering and Information Technology
TU Dortmund University
Dortmund, Germany
yagiz.sarapli@tu-dortmund.de

Abstract—Distinguishing a cyber false-data-injection attack (FDIA) from a legitimate physical disturbance, localizing it, and doing so within a protection-relevant time budget are usually studied separately. We study detection and localization across three networks – a synthetic inverter-dominated 5-bus microgrid and the standard IEEE 14-bus and 30-bus test systems – with a weighted-least-squares (WLS) AC state estimator, to test whether explicit topology-relational features help, under an evaluation enforcing strict separation between relational and non-relational feature sets, deployable WLS residuals rather than oracle state, and validation-only model selection. Against this matched, topology-free ablation, a 16-seed replication (quadrupling an original four-seed pilot) shows the gap is network-specific rather than task-specific: at both standard IEEE test systems, topology-relational features give a small, mostly statistically confirmed advantage on both detection and localization (IEEE 14-bus: $+0.009 \pm 0.012$ detection, $+0.018 \pm 0.017$ localization; IEEE 30-bus: $+0.028 \pm 0.019$ detection, $+0.013 \pm 0.016$ localization; three of the four 95% CIs exclude zero after a multiple-comparisons correction), while the small custom 5-bus microgrid shows no such advantage on either task (-0.000 ± 0.009 and $+0.003 \pm 0.017$). Separately profiling the full decision pipeline on the 5-bus implementation against a one-cycle (20 ms) budget, on a documented Apple-arm64/macOS 15.6.1 environment, gives, across two independent $N=300$ reruns, 16.2–17.0 ms median, 16.4–18.2 ms p95, and 17.0–19.2 ms p99 latency, while rare solver-tail maxima reach 89.1–92.7 ms. Neither feature set generalizes to a completely unseen attack type (0–1.3% recall, vs. 91–100% when that type is in training).

Index Terms—false data injection attack, state estimation, cyber-physical systems, attack localization, real-time systems, feature ablation, inverter-based resources

I. INTRODUCTION

Modernizing power grids toward higher penetrations of inverter-based resources (PV, battery storage, grid-forming converters) changes the character of protection and monitoring in two ways relevant here. First, inverters provide far less fault current than synchronous generators, which strains protection schemes designed around synchronous-machine fault signatures [1]. Second, the same measurement infrastructure used for state awareness is now also an attack surface: an adversary who can spoof a subset of sensor readings can attempt a false-data-injection attack (FDIA) against the state estimator that is, by construction, statistically indistinguishable from a

legitimate measurement under a residual-based bad-data test [2].

Renewable-heavy grid infrastructure specifically is not a hypothetical target: in December 2025, a destructive cyberattack – attributed to infrastructure overlapping with the Dragonfly/Static Tundra threat actor – damaged controller firmware and deployed wiper malware across more than 30 Polish wind and solar farms and a combined heat-and-power plant serving nearly half a million customers; electricity production and heat supply were ultimately not disrupted [3]. That incident was an OT/firmware-integrity attack rather than a state-estimation FDIA specifically, but it establishes the same operational stakes this paper’s threat model targets: an operator or automated scheme facing a live control-plane anomaly, under real time pressure, in exactly this class of infrastructure.

A control-room operator, or an automated protection scheme, therefore faces a three-part problem for any anomalous reading: *is this a cyberattack or a physical event, where is it*, and – if it is to inform a protective action rather than only a post-hoc forensic one – *can a decision be reached inside a protection-relevant time budget*. These three questions are usually studied separately. A large FDIA-detection literature (Section II) asks the first question, typically evaluated offline against a fixed dataset. A smaller attack-localization literature asks the second. Real-time feasibility is addressed by a much smaller, very recent body of work – notably a 2026 benchmark [4] that found deep classifiers for this exact joint task are individually fast (sub-15 ms) but the measured end-to-end pipeline is not (50–90 ms), calling this “a critical gap between algorithmic capability and protection-grade deployment” and recommending hardware acceleration. A separate, popular line of work asks whether giving a detector explicit *topology* – not just raw sensor residuals, but how each meter’s reading compares to its electrical neighbors – improves on any of this; that question is this paper’s own starting point, and its central subject.

This paper studies detection and localization within a common evaluation framework across three networks, and separately profiles real-time feasibility on the 5-bus implementation. The evaluation framework enforces strict separation

between relational and non-relational feature sets, uses deployable WLS residuals rather than oracle state, and selects models exclusively on validation data. We independently reproduce the qualitative pipeline-latency finding of concurrent work [4] (fast classifiers embedded in a slow end-to-end pipeline) and localize the actual bottleneck: not the classifier, not the state estimator’s Newton-type iteration count, but a measurement-table object being rebuilt from scratch through a general-purpose element-creation API every cycle, when its structure never changes – fixing that one thing accounts for most of the gap closed.

A. Contributions

- A network-specific result across three networks, replicated at 16 seeds after an original four-seed pilot flagged it as sign-consistent but unconfirmed: explicit topology-relational features show a small, mostly statistically confirmed detection and localization benefit at both standard IEEE test systems (14-bus, 30-bus), and no such benefit on either task at the small custom 5-bus microgrid (Section IV, Section VII) – together with a structural, quantitative redundancy analysis that rules out one natural explanation for why the two standard IEEE systems disagree with the custom microgrid.
- A root-cause profiling methodology for the “fast model, slow pipeline” gap identified in prior work, and a demonstrated fix that closes most of it without hardware acceleration (Section V).
- A further explicit negative result reported as a scope condition rather than omitted: no zero-day attack-type generalization (Section VI).

II. RELATED WORK

A. Foundational FDIA and Stealth Constructions

The FDIA threat model used throughout this paper – an attacker with knowledge of the network model constructing a measurement perturbation that leaves the WLS residual unchanged – follows Liu, Ning & Reiter’s foundational construction [2] and its AC-consistent extensions [5]. A growing subline constructs such attacks with *less* attacker knowledge than the original construction assumes: without line parameters [6], or fully “blind” to topology using generative or physics-informed models [7], [8] – all harder threat models than the one used here and natural next stress conditions for the attack generator used in this study. In the opposite direction, low-magnitude attacks close to the sensor noise floor – the specific regime Section IV’s hardened attack magnitude targets – have their own dedicated detection literature [9], worth comparing against directly in future work.

B. Deep-Learning FDIA Detection

A substantial literature applies CNN/LSTM/transfer-learning architectures to FDIA detection [10]–[12], generally evaluated as a binary classification task on a fixed attack distribution. The `residual_only` and `prior_only` baselines used here sit in this tradition; this paper’s contribution is

isolating what *explicit topology-relational* features specifically add beyond a fair combination of the two, under a stress-tested (not default-easy) attack distribution, and under features and model-selection procedures that are actually deployable rather than built with access to information a real detector would not have.

C. Topology-Aware / Graph-Based State Estimation and Detection

The closest published framing to this paper’s central question is topology-aware graph neural network (GNN) based state estimation [13]–[16], which use graph convolution/attention to exploit network structure. Prior work presents a mixed picture. [13] and an earlier, closely related study by an overlapping author group [14] compare their GNN against a same-feature “regular” DNN baseline (identical PMU-derived node features, no adjacency matrix), providing a correctly scoped topology ablation; [15]’s non-graph baseline, by contrast, does not clearly document an input-matched comparison. We therefore do *not* claim that topology ablation is generally absent from the GNN-for-power-systems literature. The contribution here is narrower: a same-architecture, same-feature comparison (GNN vs. DNN, as in [13], [14]) checks whether a different *model* benefits from topology, but it does not by itself test how the non-topological feature *set* is constructed or whether all inputs are deployable before model training – safeguards that are explicit in Section III. Section IV’s out-of-fold linear-redundancy check is offered as a second, complementary, and considerably cheaper pre-check for the feature-leakage variant of this failure mode: verifiable on the feature matrix alone, before training any model, topology-aware or not.

D. Attack/Event Localization

Localization work spans capsule networks for load-altering (not FDIA) attacks evaluated across 14/39/57-bus systems [17], moving-target-defense-assisted localization [18], and on-line detection-plus-localization schemes [19]. The cited localization studies do not explicitly document sensitivity to network topology/meshedness or an ablation of relational features against a matched non-relational baseline, which is the comparison examined in Sections IV and VII.

E. Real-Time / Latency-Aware Classification

[4] is the direct comparison point for Section V of this paper – it independently measures the same qualitative gap (fast model, slow pipeline) on a different, larger testbed using an industrial electromagnetic-transient (EMT) simulator, and calls for hardware acceleration where this paper instead finds and fixes a software-level cause for a comparable fraction of the gap.

F. Zero-Day / Unseen-Attack Generalization

The general (non-power-specific) intrusion-detection literature widely reports that supervised detectors trained on known attack families do not reliably generalize to unseen ones; a

verified, quantitative comparison specific to the power-systems FDIA literature remains an open point for future work.

G. Field-Wide Self-Assessment of Adjacent Gaps

A 2025 scoping review of ML applications in power-system protection [20] independently reports, at the level of the field rather than any single paper, two gaps this paper runs into directly: most reviewed studies do not discuss computational latency/real-time constraints at all (Section V), and cross-topology generalization is essentially untested field-wide – reviewed work concentrates on fixed IEEE 14/30/39-bus evaluations with no cross-topology testing (Section VII). This positions those sections as addressing gaps the field has already flagged in itself, not problems specific to this study’s testbed.

III. TESTBED AND METHODOLOGY

A. Network

A 5-bus, 0.4 kV synthetic microgrid built in `pandapower`: point of common coupling, a PV bus modeled as a steady-state grid-following (GFL) power injection, a BESS bus modeled as a steady-state grid-forming (GFM) power injection, and two load buses, connected by 4 lines. Both are static sgen operating-point injections, not dynamic converter-control models – no droop control or virtual inertia is simulated; the GFL/GFM labels describe each device’s intended role in the microgrid, not modeled control dynamics. `net.f_hz = 50.0` (verified from the model object, not assumed – this fixes the one-cycle protection-relevant computational target below to 20 ms).

B. State Estimation

Weighted least squares (WLS) AC state estimation via `pandapower.estimation.estimate`, with 23 measurements per cycle: voltage magnitude, and P/Q injection, at all 5 buses, plus P/Q from-side flow on all 4 lines – the same model-driven/data-driven/hybrid-driven taxonomy of state estimation under measurement anomalies surveyed in [21] implicitly underlies this paper’s own `residual_only/prior_only/topology_fusion` split. All “residual” features throughout this paper (5-bus, IEEE 14-bus, IEEE 30-bus alike) are the standard, deployable normalized measurement residual $(z - h(\hat{x}))/\sigma$ computed from the WLS solution – never a comparison to a ground-truth state a real detector could not observe.

C. Threat Model

Two synthetic attack families, both targeting a randomly chosen load bus per scenario:

- *Naive*: a single voltage-magnitude sensor is spoofed by a fixed-range random offset.
- *Model-consistent (“stealth”) FDIA*: following the Liu–Ning–Reiter construction adapted to the AC case, the true state is perturbed at the target bus and the corresponding measurement-space attack vector $a = h(x_{\text{attack}}) - h(x_{\text{true}})$

is computed exactly, so the attack is, by construction, invisible to the chi-square residual test.

A benign hard-negative physical load disturbance (a 25–70% load step at a randomly chosen bus) is included specifically to be confusable with an attack in raw residual terms.

D. Prior Construction

Every replication starts from one randomly sampled base operating point (a load/PV level, shared unmodified across that replication’s 4 scenarios). Before any attack or physical disturbance is applied to build the “true” network for a given scenario, an independent copy of that same base operating point has its load values perturbed by a fresh, independently drawn forecast error ($\mathcal{N}(0, 2.5\%)$ per load at 5-bus, clipped to $\pm 6\%$; $\mathcal{N}(0, 5\%)$ at IEEE 14-bus/30-bus, a cross-network noise-model difference documented as a limitation below) and is solved through its own, separate AC power flow – this perturbed-and-solved copy is the “prior.” `prior_only` features are the deviation between the WLS state estimate (computed from the scenario’s actual, possibly-attacked measurements) and this prior’s own voltage/angle solution. Critically, the prior is constructed *before*, and entirely independently of, whichever scenario (clean, naive attack, stealth attack, physical disturbance) that replication’s true network becomes – it never sees the true or attacked state, only an independently-perturbed forecast of the same nominal base point. This is what lets stealth attacks, invisible in residual space by construction, become visible at all: the attack is only tuned to be consistent with the WLS residual test, not with an estimate the defender never let it see.

E. Feature Sets

Four feature sets are compared. `residual_only`: conventional WLS measurement residuals. `prior_only`: the prior-deviation features defined above. `residual_plus_prior`: the two combined, with no relational or structural information at all – the topology-free ablation. `topology_fusion`: `residual_plus_prior` plus explicit relational features describing each bus relative to its network neighbors (local-minus-neighbor innovation/residual, incident-edge aggregates); no static structural metadata (node degree, device role) beyond that, since neither is a scenario-specific neighbor comparison and both risk leaking attack-target eligibility (the attack generator above only ever targets load buses). To preserve the intended ablation, non-relational baselines exclude all relational columns, and static node-degree/device-role metadata is excluded from every feature set because it could encode attack-target eligibility when attacks are restricted to load buses.

F. Models and Evaluation

LogisticRegression, RandomForest, and HistGradientBoosting classifiers are compared with replication-group-safe 70/15/15 train/validation/test splits and group-aware bootstrap confidence intervals. Model selection uses the validation split exclusively: each candidate model is scored on validation, the

TABLE I
TOPOLOGY_FUSION VS. THE CORRECTED, TOPOLOGY-FREE ABLATION
(BEST VALIDATION-SELECTED MODEL PER CELL; 5-BUS, $n=300$
DETECTION / $n=150$ CYBER-ONLY LOCALIZATION TEST SCENARIOS)

Condition	Detection (bal. acc.)		Localization (top-1)	
	topo. fusion	res.+prior	topo. fusion	res.+prior
Hard	0.950	0.937	0.967	0.967
Harder ($+2\times$ noise)	0.950	0.947	0.960	0.967

best-scoring model per feature set is selected, and only that model’s (until then unseen) test-set score is reported.

G. Sample-Size Terminology

A *replication* is one independently sampled base operating point (load level, forecast draw); each replication produces exactly 4 *scenarios* – one clean, one naive-attack, one stealth-attack, one physical-disturbance – so N replications yield $4N$ scenarios in total. The 70/15/15 split is applied at the replication level (never at the scenario level, to keep the 4 matched scenarios from one replication from leaking across train/test), so the resulting *test-scenario count* is $0.15N \times 4$, and the *cyber test-scenario count* used for localization (only the 2 attack scenario types have a real target bus) is half of that. Concretely: the headline 5-bus result uses $N=500$ replications $\rightarrow$ 2000 scenarios $\rightarrow \approx 300$ test scenarios; Section VII’s IEEE 14-bus/30-bus results use the same $N=500$ replications $\rightarrow$ 300 test scenarios for detection, 150 for localization, at both networks. Wherever this paper states “ $n=X$ ” without qualification, X is the resulting test-scenario count, not the replication count.

IV. RESULTS I: THE TOPOLOGY-FUSION ADVANTAGE DOES NOT SURVIVE ITS OWN ABLATION, AT EITHER TASK

Attack magnitude is hardened roughly $3\times$ toward the sensor noise floor, relative to the naive detection regime a first pass would produce, and evaluated at $n=300$ test scenarios (Section III). *prior_only* alone is missing residual information that *topology_fusion* has access to, so a comparison against *prior_only* specifically cannot, by construction, distinguish “topology helped” from “having residual information at all helped”; the fair comparison is against the topology-free *residual_plus_prior* ablation instead.

A. Primary Ablation

Table I compares *topology_fusion* against *residual_plus_prior* (Section III), for the primary seed and both stress conditions.

Detection shows a small edge for *topology_fusion* at this one seed, in both conditions ($+0.013$ Hard, $+0.003$ Harder); localization is an exact tie under Hard and a small edge for *residual_plus_prior* under Harder (-0.007). Neither column is bolded here – a single seed’s small, direction-inconsistent gap is not evidence of a real winner, and the four-seed replication directly below (which includes this same seed) is the test that actually matters. For reference, both feature

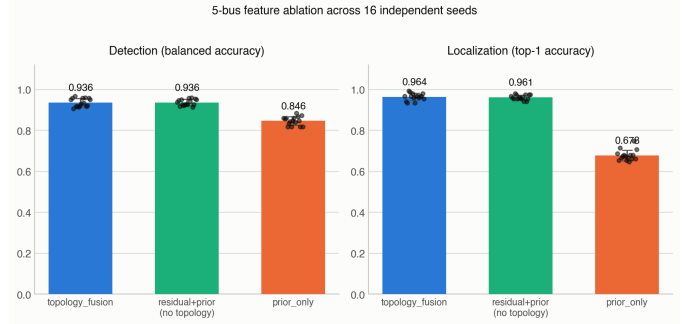


Fig. 1. Topology fusion vs. the topology-free residual+prior ablation and the prior-only baseline across 16 independent random seeds (5-bus, hard attack magnitude). Bars show mean $\pm$ standard deviation; dots show individual seed results. $n=300$ detection / $n=150$ cyber-only localization test scenarios per seed.

sets still comfortably beat *prior_only* alone (0.860–0.870 detection, 0.687–0.693 localization) and *residual_only* alone (0.700 detection, 0.547 localization) – the gain over either single baseline is large; it is simply not a topology effect.

B. Multi-Seed Replication

Figure 1 summarizes the result. Repeating full data generation and evaluation for 16 independent seeds (quadrupled from an original four-seed pilot once Section VII’s cross-network comparison called for a tighter magnitude estimate), with localization training restricted to cyber-scenario nodes only (matching the convention used at IEEE 14-bus/30-bus, Section VII), gives a *topology_fusion*-vs-*residual_plus_prior* detection gap of -0.000 ± 0.009 (range -0.013 to $+0.017$, 95% CI $[-0.005, +0.004]$) and a localization gap of $+0.003 \pm 0.017$ (range -0.020 to $+0.027$, 95% CI $[-0.006, +0.012]$): both confidence intervals cross zero, and both are, if anything, tighter around zero than the four-seed pilot’s own estimate – consistent with sampling variability around a genuinely near-zero effect, on either task.

C. Redundancy Diagnostic

Why would explicit neighbor-comparison features add nothing, given they are computed from genuinely different information – each bus’s residual/innovation relative to its neighbors, not just its own? On this fixed, small (5-bus, 4-line) topology, the result is consistent with high linear recoverability of the relational features from the local feature set. Predicting each of the 65 relational columns from *residual_plus_prior* alone, with 5-fold cross-validated ridge regression, gives a mean out-of-fold R^2 of 0.956 (median 1.000, minimum 0.613, 90.8% of columns above 0.8): a neighbor-mean or incident-edge aggregate over a fixed, small neighbor set is, on a network this size, close to an affine function of the same per-bus values *residual_plus_prior* already provides in full. A sufficiently flexible classifier (Hist-GradientBoosting, the best model throughout) may therefore reconstruct much of that relational information internally without ever being given it explicitly, which is exactly consistent

with Table I’s near-ties on both tasks. This explanation is specific to a fixed, small topology, and does not by itself predict what happens at larger, standard networks (Section VII), where an unrelated, size-agnostic feature design is used instead – reduced recoverability of relational features from local information remains one possible condition under which explicit topology-awareness might help; Section VII tests whether such benefits appear on larger networks, while the Discussion examines whether reduced redundancy explains the observed pattern.

V. RESULTS II: REAL-TIME FEASIBILITY – DIAGNOSING AND FIXING THE GAP

Following [4]’s finding that fast classifiers can still sit inside a slow end-to-end pipeline, we measured the full per-cycle decision pipeline (state estimation → feature computation → classification) against a one-cycle (20 ms at the network’s confirmed 50 Hz) protection-relevant computational target – not a claim that every protective relaying function requires exactly this latency, but the natural reference point for a per-cycle decision loop – with one-time network/model setup excluded, matching how a deployed scheme would actually operate.

A. Initial Measurement

Median 42.98 ms, $2.15\times$ over budget, using the original pipeline (RandomForest classifier, tolerance=1e-7, maximum_iterations=40, flat state-estimator initialization every cycle).

B. Warm-Start Test

Warm-starting the state estimator from the previous cycle’s converged solution produced only a $1.06\times$ speedup (37.14 ms), ruling out “far from the solution” as the dominant cost.

C. Profiling

Splitting the state-estimation call into its constituent steps showed the measurement-submission step cost 19.9 ms – essentially the same as the estimator’s own solve (19.4 ms). An ablation on maximum_iterations (40 → 5: no speed change; 40 → 3: convergence failure in 19/20 trials) showed the true iteration requirement is roughly 4–5 steps regardless of the 40-iteration cap, ruling out iteration count as the estimator-side cost driver.

D. Measurement-Update Optimization

The measurement table was being rebuilt from scratch every cycle via a general-purpose element-creation API, called once per channel in a Python loop – even though the table’s structure is identical every cycle; only the values change. Replacing this with a one-time table construction followed by an in-place overwrite of the value column each cycle gave a $1158\times$ speedup on this sub-step (19.4 ms → 0.017 ms), verified to produce a bit-identical state-estimation result (max difference 0.0).

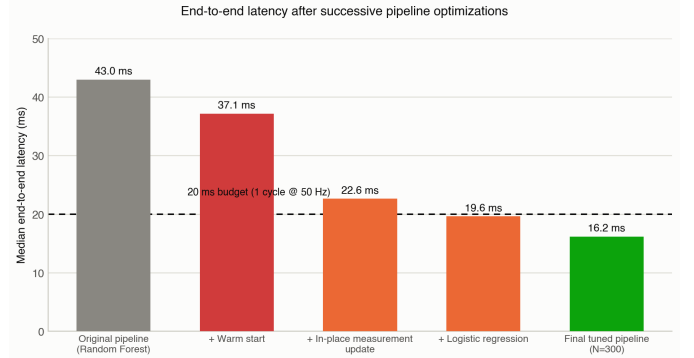


Fig. 2. End-to-end latency after successive pipeline optimizations. In-place measurement updates account for most of the reduction; the lighter classifier provides a smaller additional improvement.

E. Final Benchmark

With a lighter classifier (LogisticRegression, competitive with RandomForest per Section IV’s accuracy results and faster at inference), two independent $N=300$ reruns on the same documented environment, both with 0 convergence failures, give 16.155–17.015 ms median, 16.429–18.211 ms p95, 16.952–19.190 ms p99, and 89.067–92.657 ms max end-to-end latency (Fig. 2 shows the most recent run). Median, p95, and p99 remain within the 20 ms computational budget in both reruns, while the maximum exceeds it in both. Across the same two reruns, stage medians are 15.855–16.664 ms for WLS state estimation, 0.050–0.057 ms for post-estimation residual/innovation calculation, 0.083–0.088 ms for feature computation, 0.0068–0.0076 ms for array assembly, and 0.160–0.200 ms for model inference; the WLS solve therefore dominates the budget. The benchmark environment is Apple arm64 running macOS 15.6.1 with Python 3.12.4, pandas 2.14.10, NumPy 1.26.4, pandas 2.2.2, and scikit-learn 1.4.2. This benchmark uses its own, separately implemented 35-feature representation (7 local-plus-neighbor statistics per bus $\times$ 5 buses) built for the timing measurement specifically, not topology_fusion or residual_plus_prior verbatim; feature computation is a small fraction of the total either way, so the exact feature set used for classification is a second-order cost driver. The classifier is trained on 150 real warm-up scenarios generated through the same simulated pipeline (83 attack, 67 clean; training balanced accuracy 1.000), and the residual/innovation features use the same deployable quantities as everywhere else in this paper; the timer covers all five pipeline stages end to end (state estimation, post-estimation residual/innovation calculation, feature computation, array assembly, inference), so the headline numbers above are the true total, nothing excluded.

F. Positioning

[4] measures the same category of gap (sub-15 ms classification embedded in 50–90 ms pipelines) on an industrial-grade EMT simulator and recommends hardware acceleration. This paper’s finding – that a large fraction of a comparable

TABLE II
RECALL ON A WITHHELD VS. TRAINED-ON ATTACK TYPE ($n=75$ TEST SCENARIOS PER CELL, BOTH CONDITIONS)

Attack type	Features	Withheld	In training
Naive corruption	topo. fusion	0.000	0.907
Naive corruption	res.+prior	0.000	0.907
Stealth FDIA	topo. fusion	0.013	1.000
Stealth FDIA	res.+prior	0.013	1.000

gap traced to avoidable software overhead rather than model inference – suggests pipeline profiling, not only hardware acceleration, deserves consideration first. A second, independent data point supports the “classifier inference itself is cheap” half of this argument: a knowledge-distilled LightGBM detector for virtualized microgrids reports 54–67 ms per 1000 samples (~ 0.06 ms/sample) for inference alone [22] – the same order of magnitude as this paper’s own model-inference sub-step (~ 0.16 – 0.20 ms/sample), and far below this paper’s state-estimation sub-step (~ 15.9 – 16.7 ms/sample). Both results point the same way: once a classifier is lightened, the remaining latency budget is spent upstream of the model, not in it.

VI. RESULTS III: THE ADVANTAGE DOES NOT EXTRAPOLATE TO UNSEEN ATTACK TYPES

To test whether Section IV’s detector generalizes beyond the two attack families it was trained on (as opposed to merely discriminating them well), we trained HistGradientBoosting – with the same tuned hyperparameters used as the reported 5-bus detector elsewhere in this paper – with one entire attack family withheld from training and validation, then measured recall on exactly that withheld family, evaluated against the standard test split under a symmetric protocol (the held-out and seen-in-training conditions are compared on the same kind of row) (Table II, Fig. 3). Run for both `topology_fusion` and the topology-free `residual_plus_prior` ablation, to check directly whether the collapse below is specific to one feature set or, as Section IV’s in-distribution finding would predict, behaviorally identical for both.

The detector recognizes the statistical signature of attack types represented in its training data – including the stealthy one, very well – but does not extrapolate to a mechanism it has not seen, and this holds identically whether or not topology-relational features are included: the two feature sets are indistinguishable on every cell of Table II, confirming Section IV’s prediction that they are behaviorally interchangeable, here for a question – generalization to an unseen mechanism – that has nothing to do with topology specifically. We report this as a hard scope limit on any deployment claim built on Section IV’s results: within this study’s two-family attack taxonomy, this is a complete absence of zero-day capability, not a partial one, for either feature set. This complements, from the opposite direction, a related generalization gap documented in the fault-classification literature: an XGBoost-based classifier trained and evaluated on the *same* distribution still

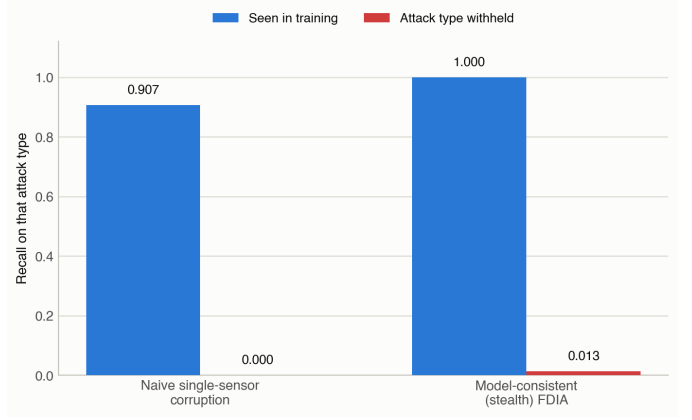


Fig. 3. Recall for attack types represented in training versus withheld from training (topology fusion shown; residual+prior is behaviorally identical, Table II).

drops from 96.8% (synthetic data) to 85.2% (physical data) [23] – i.e., even without an unseen attack *type*, a synthetic-to-real distribution shift alone costs over 11 points. Between that gap and this paper’s own result, generalization – of any kind – looks like the open problem for this class of detector, not an edge case. A deployment-aware zero-day evaluation in the adjacent EV/V2X charging-infrastructure domain [24] is a useful methodological reference for what a more rigorous version of Section VI’s test could look like.

VII. RESULTS IV: CROSS-NETWORK EVALUATION REVEALS TASK- AND NETWORK-SPECIFIC EFFECTS

Everything in Sections IV–VI uses the custom 5-bus microgrid. We separately tested whether explicit topology-relational features help at a larger, standard scale, using a completely independent, size-agnostic feature-engineering design.

A. Test-System Selection

IEEE 14-bus and IEEE 30-bus were selected because their pandapower bus indexing maps directly to the internal power-flow representation used by the evaluation pipeline (confirmed directly: `net.bus` count matches the internal power-flow-case bus count for both), and they are, independently, among the most commonly used benchmarks in the power-system state-estimation/FDIA literature specifically.

B. Cross-Network Method

All WLS/measurement/attack machinery is network-size-agnostic by construction and was reused unmodified. The original fixed per-bus feature columns do not generalize past 5 buses and were replaced with a size-agnostic design: global summary statistics of the residual/innovation arrays, plus, for localization, a trained per-node classifier ranking every bus’s probability of being the attack target. This feature design is implemented separately from Section III’s but applies the same safeguards: the residual feature is the deployable measurement residual $z - h(\hat{x})$ (never the simulator’s hidden ground-truth

TABLE III

5-BUS VS. IEEE 14-BUS VS. IEEE 30-BUS, ALL 16-SEED MEANS, AT MATCHED TEST-SET SIZES ($n=300$ DETECTION / $n=150$ CYBER-ONLY LOCALIZATION SCENARIOS PER NETWORK)

	topo.	res.+prior	prior	res.
Det., 5-bus	0.936	0.936	0.846	0.680
Det., IEEE-14	0.882	0.873	0.705	0.663
Det., IEEE-30	0.877	0.849	0.680	0.646
Loc., 5-bus	0.964	0.961	0.678	0.568
Loc., IEEE-14	0.925	0.907	0.531	0.462
Loc., IEEE-30	0.901	0.888	0.510	0.430

Per-seed gap direction and uncertainty are reported separately in Section IV and Table IV; no cell is emphasized as a confirmed winner.

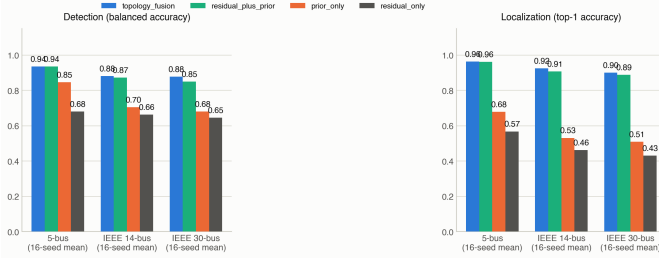


Fig. 4. 5-bus vs. IEEE 14-bus vs. IEEE 30-bus, all 16-seed means, all four feature sets. The gap is network-specific, not task-specific: null on both tasks at 5-bus, a small but mostly statistically confirmed advantage on both tasks at IEEE 14-bus and IEEE 30-bus (Table IV).

state), model selection is validation-based rather than test-set-based, non-relational baselines exclude relational columns, and *topology_fusion* excludes static structural metadata such as node degree, which could otherwise act as a target-eligibility prior because attacks are restricted to load buses. All numbers below are from this corrected pipeline, re-run in full at $N=500$ replications, 16 seeds (quadrupled from an original four-seed pilot; Section VII-D explains why), at both networks.

C. Cross-Network Result

At 5-bus (Table IV), both gaps remain null at 16 seeds, essentially unchanged from the original four-seed pilot (Section IV) but with a visibly tighter interval around zero. At IEEE 14-bus, the detection gap is positive in 11 of 15 non-zero seeds (mean $+0.009 \pm 0.012$) and the localization gap is non-negative in all 16 seeds, zero seeds favoring *residual_plus_prior* (mean $+0.018 \pm 0.017$). At IEEE 30-bus, the detection gap is positive in all 16 seeds (mean $+0.028 \pm 0.019$) and the localization gap is positive in 12 of 15 non-zero seeds (mean $+0.013 \pm 0.016$). Unlike the four-seed pilot, in which each IEEE network appeared to favor a single, different task, both tasks now show a positive gap at both standard IEEE networks.

D. Statistical Confirmation at 16 Seeds

An original four-seed pilot found every gap’s 95% confidence interval crossing zero and flagged the pattern as sign-consistent, not statistically confirmed, explicitly calling for roughly 10–20 seeds to resolve the effects’ magnitude.

TABLE IV

TOPOLOGY_FUSION VS. RESIDUAL_PLUS_PRIOR (IEEE-14/30: THE STRONGEST NON-RELATIONAL BASELINE IN EVERY SEED), 16 INDEPENDENT SEEDS (500 REPL. EACH), ALL THREE NETWORKS

Network	Task	Mean gap	95% CI	Sign p	t -test p
5-bus	Det.	-0.000 ± 0.009	$[-0.005, +0.004]$	0.500	0.849
5-bus	Loc.	$+0.003 \pm 0.017$	$[-0.006, +0.012]$	0.212	0.500
IEEE-14	Det.	$+0.009 \pm 0.012$	$[+0.002, +0.016]$	0.059	0.011
IEEE-14	Loc.	$+0.018 \pm 0.017$	$[+0.008, +0.027]$	<0.001	<0.001
IEEE-30	Det.	$+0.028 \pm 0.019$	$[+0.018, +0.038]$	<0.001	<0.001
IEEE-30	Loc.	$+0.013 \pm 0.016$	$[+0.004, +0.022]$	0.018	0.007

Sign test excludes exact ties. A Bonferroni correction for 6 simultaneous tests needs $p < 0.0083$ (family-wise 95%); three of the four positive gaps clear it; IEEE-14 detection ($p = 0.011$) does not.

Quadrupling to 16 seeds does resolve it, for most of the six gaps (Table IV): four now have a t -based 95% confidence interval excluding zero, with a two-sided t -test rejecting the null at $p < 0.05$ for the same four (IEEE-14 detection $p = 0.011$; IEEE-14 localization $p < 0.001$; IEEE-30 detection $p < 0.001$; IEEE-30 localization $p = 0.007$). A distribution-free sign test, weaker but assumption-free, agrees for three of the four (IEEE-14 localization $p < 0.001$; IEEE-30 detection $p < 0.001$; IEEE-30 localization $p = 0.018$) but not for IEEE-14 detection specifically ($p = 0.059$, 11 of 15 non-zero seeds positive) – evidence that a small effect relative to per-seed noise can clear a parametric test before it clears the sign test’s coarser bar. Testing six gaps simultaneously also raises a multiple-comparisons concern: a Bonferroni correction requiring $p < 0.0083$ for family-wise 95% confidence still clears IEEE-14 localization, IEEE-30 detection, and IEEE-30 localization, but not IEEE-14 detection. We therefore report three of the six gaps (IEEE-14 localization, IEEE-30 detection, IEEE-30 localization) as confirmed at a conservative, multiple-comparisons-aware threshold, IEEE-14 detection as directionally consistent but the weakest-supported of the four positive gaps, and both 5-bus gaps as clearly null ($p = 0.849$ and $p = 0.500$ by t -test). Sixteen seeds remains a convenience choice rather than a formal power calculation; a pre-registered sample size or a paired/bootstrap difference test, neither run here, would put a tighter, less assumption-dependent bound on the confirmed effects’ true magnitude.

E. Interpretation

Across the three networks and two tasks, the 16-seed pattern is network-specific rather than task-specific: both detection and localization show a small, mostly confirmed positive gap at both standard IEEE test systems, while the small custom 5-bus microgrid shows no such gap on either task. This differs from the four-seed pilot, whose smaller sample made each IEEE network appear to favor a single, different task – the four-seed IEEE-14 detection mean (-0.003) sat well within the 16-seed distribution’s own spread (std 0.012), so the apparent task-network pairing was itself within ordinary small-sample noise, exactly the possibility the pilot’s own statistical caveat had flagged. This is consistent with, and sharpens, a 2025 field-wide scoping review’s [20] independent finding that cross-

topology generalization is essentially untested across the ML-for-power-protection literature it surveyed – this study is a direct, sample-size-matched, cross-implementation attempt at a gap the field has already named, carried to a seed count the reviewed literature typically does not report at all.

VIII. DISCUSSION

Read together, Sections IV–VII converge on a pattern that is network-specific rather than task-specific, once the seed count is large enough to separate a genuinely small effect from per-seed noise (Section VII-D). At the small custom 5-bus microgrid, explicit topology-relational features show no advantage over a matched non-relational ablation on either detection (mean -0.000 ± 0.009 , 16 seeds) or localization (mean $+0.003 \pm 0.017$). At both standard IEEE test systems, the same ablation shows a small, mostly confirmed positive gap on both tasks: IEEE 14-bus detection ($+0.009 \pm 0.012$) and localization ($+0.018 \pm 0.017$); IEEE 30-bus detection ($+0.028 \pm 0.019$) and localization ($+0.013 \pm 0.016$). Put plainly: the two standard IEEE test systems behave alike, and differently from the small custom microgrid, on both tasks – not, as an original four-seed pilot appeared to show, a different single task favored at each network.

What helps detection at 5-bus, considerably, is combining residual and prior-innovation information: a consistent multi-seed gain over either source alone (Section IV) – not a topology-specific effect. Section IV’s redundancy analysis gives the 5-bus null result a structural, not just empirical, footing: on a fixed, small topology, a relational aggregate is close to a linear function of information a flexible classifier already has (mean out-of-fold $R^2 = 0.956$), so there is little left for an explicit relational feature to contribute. If this redundancy argument explained the cross-network pattern, it should weaken at the larger, more meshed IEEE networks – but the same diagnostic, applied to IEEE-14’s relational localization features, gives a mean R^2 of 0.966 (median 0.999, 92% of columns above 0.8), if anything *higher* than 5-bus, not lower. So the redundancy hypothesis is not simply confirmed either: the relational information stays highly linearly recoverable from local features at IEEE-14, and giving it explicitly still measurably helps, now on firmer statistical footing than the four-seed pilot allowed. Both facts can be true together – a classifier being *able* to reconstruct information linearly does not mean a fixed-depth, greedily-built ensemble reliably does, especially without ever being offered the aggregation as a single feature – but this diagnostic alone does not resolve which is happening; it only rules out “the information isn’t linearly there” as the explanation. We have not run the equivalent check for IEEE-30’s detection-relevant relational features, for the same representation-mismatch reason as before: IEEE-30’s detection features and 5-bus’s use different, non-comparable representations, and building a third variant of this diagnostic was judged out of scope here.

One plausible explanation, though not directly verified, is discoverability rather than availability: HistGradientBoosting – a tree-based model with no built-in notion of “neighbor”

– must find the right subset of a flat, unstructured input to combine for each target node, a search that may be harder on a larger, more complex network even when the information needed is linearly present throughout; being handed the relational feature directly removes that search entirely. Unlike the four-seed pilot’s pattern (one task benefiting at IEEE-14, a different task at IEEE-30, no shared story to connect them), a discoverability account that scales with network size/complexity rather than switching unpredictably between tasks is at least directionally consistent with the 16-seed result: both standard IEEE networks, larger and more meshed than the custom 5-bus microgrid, now show the same qualitative pattern on both tasks. This remains a genuine open question this paper raises rather than resolves – the mechanism linking network scale or topology to when explicit relational features help is not identified here, only the pattern itself, on firmer statistical footing than the four-seed pilot allowed.

The 16-seed replication therefore resolves what the four-seed pilot’s own stated caveat flagged as unconfirmed: the gap is small, but real and tied to network identity – present at both standard IEEE test systems on both tasks, absent at the small custom microgrid on both tasks – rather than a coincidental, task-specific pattern with no shared explanation.

IX. LIMITATIONS

- 1) **Scale and network-specificity:** Three network topologies have now been tested to the same 16-seed standard (quadrupled from an original four-seed pilot specifically to resolve this cross-network pattern’s statistical confirmation, Section VII-D). Both tasks now show a small, mostly confirmed positive gap at both standard IEEE systems (14-bus, 30-bus); neither task shows any gap at the small custom 5-bus microgrid. None uses a real (non-synthetic) network or load profile, and with only three networks tested, one custom and two standard, the finding that the two standard networks behave alike and differently from the custom one is consistent with, but does not prove, a genuine standard-vs-custom or scale/meshedness effect – a fourth network, ideally another custom, non-standard design, would be needed to separate “standard IEEE benchmark” from “larger or more meshed” as the operative variable.
- 2) **Multiple comparisons and one borderline result:** four of the six gaps in Table IV have a 95% confidence interval excluding zero and $p < 0.05$ by t -test; a Bonferroni correction for six simultaneous tests (requiring $p < 0.0083$ for family-wise 95% confidence) leaves three of four surviving – IEEE-14 detection ($p = 0.011$ uncorrected) is the weakest-supported of the four and does not clear the sign test’s conventional threshold either ($p = 0.059$). Even where confirmed, effect sizes remain small (0.009–0.028 balanced-accuracy/top-1 points) relative to the much larger gain from combining residual and prior information over either alone (Section IV). Sixteen seeds is itself a convenience choice, not a formal power calculation; a pre-registered sample size or a

paired/bootstrap difference test, neither run here, would put a tighter, less assumption-dependent bound on the confirmed effects’ true magnitude.

- 3) **Cross-network measurement-noise and forecast-uncertainty mismatch:** measurement standard deviations are fixed absolute values shared, unscaled, across all three networks despite their very different physical scales, and prior load-forecast uncertainty is 2.5% at 5-bus vs. 5% at IEEE-14/IEEE-30 (Section III). This was not standardized or swept as an independent robustness axis; the cross-network comparisons in Section VII should be read with this confound explicitly in mind, not as a controlled scale-only experiment. In light of this, “network-specific” throughout this paper is more precisely *setup-and-network-specific under the tested measurement/prior regimes*: each network’s result reflects its topology together with its own noise/prior configuration, not topology in isolation. IEEE-14 and IEEE-30 share an identical noise/prior convention, so their disagreement with each other (one task’s gap each, not the same task) is not subject to this particular confound; only comparisons that include 5-bus are.
- 4) **No zero-day generalization** (Section VI): detection/localization numbers hold only for attack types represented in training – verified for both `topology_fusion` and `residual_plus_prior`, which collapse identically (Table II), under a corrected, symmetric evaluation protocol; localization was not separately re-tested for zero-day generalization (only detection was), and the underlying attack taxonomy is still limited to two families (item 5 below).
- 5) **Single testbed family, two attack types:** the present AC-consistent attack generator does not independently vary the compromised-measurement subset; introducing explicit attacker-access constraints (e.g. a fixed budget of compromisable sensors) as a separate, tunable threat-model parameter is left for future work.
- 6) **Latency results** (Section V) characterize one documented Apple-arm64/macOS 15.6.1 Python/pandapower environment, not a hardware or RTOS claim, and were not re-benchmarked on the smaller `residual_plus_prior` feature set (Section V argues this would not change the conclusion, since state estimation, not classification, dominates the budget, but this is an argument, not a re-measurement).

X. CONCLUSION

Across a small custom inverter-dominated microgrid and two standard, larger IEEE test systems (14-bus and 30-bus), explicit topology-relational features show a network-specific, not task-specific, pattern. At 16 seeds, quadrupled from an original four-seed pilot specifically to resolve this pattern’s statistical confirmation, both detection and localization show a small but mostly confirmed positive gap at both standard IEEE systems (IEEE-14: $+0.009 \pm 0.012$ detection, $+0.018 \pm 0.017$ localization; IEEE-30: $+0.028 \pm 0.019$ detection, $+0.013 \pm$

0.016 localization), while the small custom 5-bus microgrid shows no such gap on either task. This differs from the four-seed pilot, whose smaller sample made each IEEE network appear to favor a single, different task – a pattern this paper’s own statistical caveat had already flagged as consistent-sign but not confirmed, and which the larger seed count has since resolved in the more coherent, network-specific direction. We take the remaining question – *why* standard IEEE-scale networks show this advantage and the custom microgrid does not – as a genuine one this paper raises rather than settles (Section VII), and a caution against generalizing any single network’s result into a general rule from too few data points, of either networks or seeds. Separately, the resulting decision pipeline, verified with a trained classifier and deployable features throughout, can be brought within a one-cycle target once its actual bottleneck, a software inefficiency, is fixed, and the detector does not extend to attack types unseen in training.

CODE AND DATA AVAILABILITY

Code and reproducibility materials are available at <https://github.com/yagizsarapli/cyber-physical-grid-detection>.

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